

साक्षी

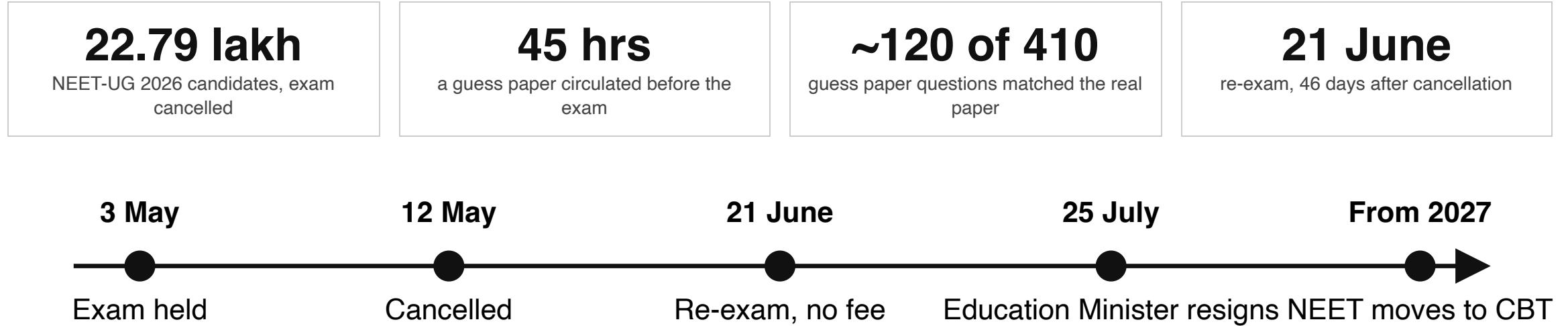
Saakshi

A resilient, provable computer-based testing ecosystem

Hackathon Challenge 6 · Problem Statement & Proposed Solution

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What really happened in India



Saakshi is a computer exam system where every answer is signed, every failure is survived and every decision can be checked.

Four real incidents

NEET-UG 2026: a leak at the source

Insiders at a printing press scanned papers before dispatch. The leak spread to more than 10 states. The CBI made arrests. The leak came from the people who handle the paper, not from the exam hall.

NEET-UG 2024: unfair fixes

1,563 candidates got grace marks for lost time. The marks were scrapped and a re-test was offered. Only 813 took it. The court noted wrong paper sets and weak CCTV.

SSC 2025: the insider at the centre

Exams were cancelled after crashes and biometric failures. At Dhanbad the server manager was arrested. Answers were auto-selected while candidates only moved the mouse.

JEE Main 2025: no proof for the candidate

A pop-up showed 46 questions attempted. The response sheet showed 29. Candidates can challenge the answer key, but not their recorded responses.

Why now: about 22 lakh candidates will sit NEET on computers from 2027. The system they use must be ready before that.

Sources: Wikipedia "2026 NEET controversy"; Outlook India; Gulf News and SCC Online; MedicalDialogues; SCC Online 2 Aug 2024; Careers360 (SSC Phase 13, SSC CGL Dhanbad, JEE Main 2025); Bihar CHO exam report, All India Radio. Full list in docs/research.md.

The problem, broken into six failure types

Each failure type needs a different answer. One fix does not cover all of them.

1. Paper leak at the source

Seen in: NEET-UG 2026

Someone who sets or prints the paper shares it.

Saakshi: papers stay locked until two of three key holders release them. It cannot stop a setter who knows the questions.

2. Insider changes answers

Seen in: SSC Dhanbad, Bihar CHO (37 arrests)

A centre server or its operator edits or injects responses.

Saakshi: each answer is signed on the seat and chained. Any later edit is found and the true answer is recovered.

3. Outage and crashes

Seen in: CUET-UG 2026 (3,765 candidates re-tested), SSC 2025

Software or links fail during the exam.

Saakshi: the seat keeps answers on its own disk. Killed relays and cells were rebuilt with 0 answers lost.

4. Remote help and cheating tools

Seen in: SSC 2025 remote access, Bihar CHO proxy server

Remote-view software or hidden tools run on the exam PC.

Saakshi: a check before the exam blocks known remote tools by name. Flags go to a human, never an automatic penalty.

5. No proof for the candidate

Seen in: JEE Main 2025 (46 vs 29)

The candidate cannot check what was recorded.

Saakshi: each candidate gets a receipt slip. The public verify page compares it with the record.

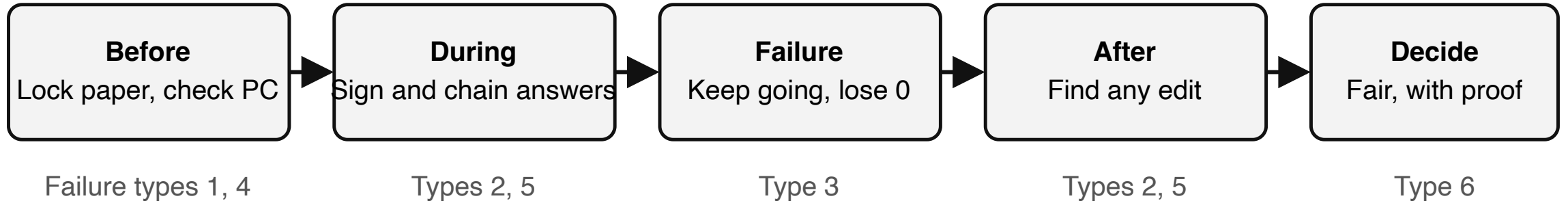
6. Unfair handling after trouble

Seen in: NEET-UG 2024 (grace marks, 813 of 1,563 re-tested)

Who gets compensated, re-tested or re-conducted is argued in court.

Saakshi: a rule-based decision per candidate and per centre, shown with evidence. Tested on synthetic data only.

How Saakshi answers them, in one picture



Incidents: Careers360 (SSC, JEE Main 2025); Business Standard (CUET-UG 2026 re-test); All India Radio (Bihar CHO); SCC Online (NEET-UG 2024). Saakshi results: docs/evidence/ (chaos drill 20 of 20 passed, 16,000 answers sent and 16,000 stored; tamper lab found 1,000 of 1,000 byte flips).

Does it fit? The 12 focus areas (1 to 6)

7 of 12

built and tested

5 of 12

partly built, rest named below

0 of 12

not started

#	Focus area	What Saakshi does	Fit
1	Real-time monitoring	Live seat grid and control-room tiles per centre. A silent seat shows after 30 seconds.	Partly built
2	Early failure prediction	Risk score per centre: allot, add an observer, or do not allot, with the top 3 reasons. Synthetic drill data only.	Partly built
3	Incident detection and escalation	Incidents ranked by severity, with an escalation ladder, a regulator report draft and invigilator reports sorted by type.	Built and tested
4	Backup and disaster recovery	Relays store and forward. A wiped cell is rebuilt from relays. 3 cells run together. Lost answers: 0.	Partly built
5	Tamper-evident storage	Signed, chained answers. A per-shift register that an independent witness checks. Keys split 2 of 3.	Built and tested
6	Suspicious patterns	Pattern signals on synthetic data: 53 flags, no honest candidate flagged. A human always reviews.	Partly built

Does it fit? The 12 focus areas (7 to 12)

#	Focus area	What Saakshi does	Fit
7	Reconciliation and validation	Missing answers are requested again. Duplicates are ignored. Each centre-shift report ties registered, submitted and receipts together.	Built and tested
8	Candidate communication	Tick states, offline message and receipt slip in English and Hindi. Approved notices in English, Hindi and Tamil.	Built and tested
9	Re-schedule or re-conduct decision	Per-candidate compensate or re-test, plus centre-level re-conduct. Checked on NEET-UG 2024 and CUET 2026 cases.	Built and tested
10	Fairness when disrupted	Rules that treat candidates alike. Time-gap records and caps on compensation are still to build.	Partly built
11	Audit trail and evidence reports	A signed evidence pack anyone can verify. The verify page shows the exact changed answer.	Built and tested
12	AI analytics for systemic risk	Risk scores, incident sorting and pattern signals on the cells' own export.	Built and tested

What it does well, and honest limits

What the solution does well

It removes the need to trust any one insider. It keeps the exam running through failures. It gives the candidate proof.

Honest limits

A seat signature shows a record was not changed after the device made it. It does not show the candidate chose it. It cannot stop a paper setter from leaking questions. Analytics use synthetic data. Speed figures come from one machine. Tested on macOS and Windows.

Repository: <https://github.com/anirudh12032008/saakshi>